

TWINKIT · DESIGN PRECIS

TwinKit design precis

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
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REVISION HISTORY

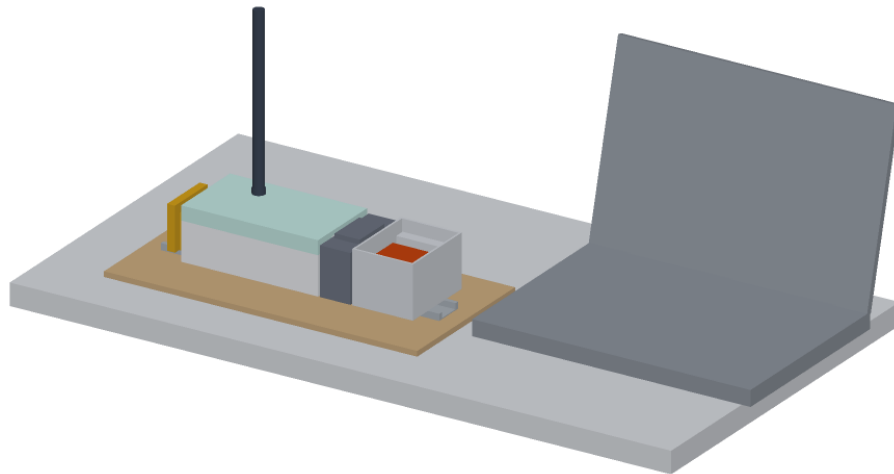
Version	Date	Author	Change
0.4	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002). Adaptive data rate and cool-hours job scheduling become design rules; \$300 budget
0.3	2026-09-25	Amish Chadha	TRL 3 sizing results from TWK-CAL-001, design choices adopted per TWK-DDR-001, fuse and UPS charger changes
0.2	2026-09-25	Amish Chadha	Populate to TRL 2 (architecture, first-order numbers, safety, media)
0.1	2026-09-25	Amish Chadha	Initial scaffold

TwinKit design precis

TwinKit is a small DIN rail edge gateway plus open software that receives sensor readings, stores them, and shows each reading on the project's own build123d model next to the value the design calculation predicted. The TRL 3 calculations (TWK-CAL-001) show one single-board computer with a LoRaWAN concentrator serving 50 sensor nodes, storing a year of data in 4.7 GB and riding through a 2.4 h outage on 6.4 W. With adaptive data rate, uplink loss is 0.21 % against 1 % (R1), and the \$290 parts cost fits the \$300 budget Amish set on 2026-09-25 (R14). The processor stays 15.9 K below its assumed throttle point at 40 °C and light load, with heavy jobs moved to cool hours (R12). The design choices below were proposed at TRL 2 and decided by Amish on 2026-09-25 (TWK-DDR-001, TWK-DDR-002).

TwinKit

CONCEPT, NOT FOR FABRICATION



Grey: Desk, 14-inch laptop for scale

Figure 1. Gateway on a DIN rail and bench plate, with a desk and 14-inch laptop for scale.

■ How it works

1. **Sense.** Sensor nodes such as FieldNode (a sibling lab project) send small readings over LoRa radio. Other devices can post over MQTT or HTTP on the local network.
2. **Receive.** An 8-channel LoRaWAN concentrator on the gateway hears all nodes in range; an open-source network server decodes the packets and publishes them to a local MQTT broker.

3. **Store.** A time-series database on the gateway keeps every reading with its node, channel, unit and time.
4. **Map.** A mapping file for each project (the twin file) links every channel to a named part of the project's build123d model, using the same part names and BOM numbers the concept media already use, and to an expected value: a constant, a curve, or a function from the project's calculation note.
5. **Compare.** The TwinKit twin service computes the residual (measured minus expected) for each reading and raises a flag when it stays outside the stated tolerance for the stated time.
6. **Show.** A browser dashboard shows trends and flags, and a 3D view loads the project's `model.glb` and colors each mapped part by its live value or residual.

TwinKit: data flow from sensor to twin (values are TRL 3 estimates, TWK-CAL-001)

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Figure 2. Data flow from sensor to twin. Values are estimates.

An illustrative twin file for the first example (FieldNode) reads:

```

project: fieldnode
model: media/model.glb
channels:
  - id: battery_v
    node: fieldnode-01
    part: 4          # BOM number of the LiFeP04 cell in FieldNode's model
    unit: V
    expected: calc.power_budget.soc_to_voltage  # from FieldNode's calculation note
    tolerance: 0.1
    hold: 30 min
  
```

The schema fields (project, model, and per channel an id, node, part, unit, expected value, tolerance and hold time) were decided by Amish on 2026-09-25 (TWK-DDR-002). The hold time sets how fast a fault is flagged: with a 30 min hold, a step fault is flagged within 35 min at 5 min readings (TWK-CAL-001 [C2]). The twin service commits readings in batches every 10 s to limit card wear (TWK-CAL-001, section B).

Two operating rules follow from Amish's decisions of 2026-09-25 (TWK-DDR-002) and are part of the design:

- **Adaptive data rate on.** The network server runs LoRaWAN adaptive data rate, so each node settles on the fastest spreading factor its link allows (SF7 to SF9 in a lab or building). This keeps collision loss at 0.21 % for 50 nodes at 5 min (TWK-CAL-001 [A6]); with every node at SF9 it would be 1.02 %. FieldNode nodes must accept adaptive data rate commands.
- **Heavy jobs in cool hours.** Database recompression, backups and dashboard rebuilds are scheduled by the twin service for the coolest hours (by default 02:00 to 05:00 local, or when the board temperature is below a set value), so that the processor runs at light load when the room is warm (TWK-CAL-001 [F6]).

■ Main components

Table 1. Main components (numbers match `bom/bom.csv` and Figure 3).

#	Component	Choice	Notes
1, 2	Bench plate and DIN rail	6 mm plate, TS35 rail	Plate optional when an equipment cabinet exists
3, 4	Gateway enclosure	9-module DIN enclosure, vented, clear cover	Holds the computer and concentrator
5, 6	Single-board computer and cooler	4 GB quad-core Arm board (Raspberry Pi 5 class)	Runs all software; about 2.4 GB used. Decided by Amish, 2026-09-25 (TWK-DDR-002)
7	LoRaWAN concentrator	SX1302 or SX1303 8-channel HAT	Region version (EU868 or US915). Decided by Amish, 2026-09-25 (TWK-DDR-002)
8	Storage	64 GB high-endurance microSD	SSD upgrade is an option
9	Antenna	3 dBi whip on an SMA bulkhead	Mounted on the cover or outside a cabinet; tip 264 mm above the plate
10	DC-DC converter	9 to 30 V in, 5.1 V 5 A out	Accepts a 12 V adapter or an existing 12 V system
11, 12	Backup	DIN UPS module with a 9 to 30 V buck-boost charger and a 12.8 V, 1.5 Ah LiFePO4 pack with built-in BMS	Power-fail signal triggers a clean shutdown. Pack choice decided by Amish, 2026-09-25 (TWK-DDR-002)
13	Terminals and fuse	DIN terminal blocks, 3.15 A time-delay fuse	Fuse at the supply input; raised from 2 A by TWK-CAL-001 [D3]
14	Open-source stack	LoRaWAN network server, MQTT broker, TimescaleDB, dashboard	All open source, no cloud account. Decided by Amish, 2026-09-25 (TWK-DDR-002)
15	Twin service and schema	Python service and YAML twin file	TwinKit's own code, MIT
16	Example twin	FieldNode battery and solar charge	Decided by Amish, 2026-09-25 (TWK-DDR-002)

TwinKit: exploded view

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- 1 Bench mounting plate
- 2 DIN rail, TS35
- 3 Gateway enclosure base
- 4 Gateway enclosure cover
- 5 Single-board computer, 4 GB
- 6 Active cooler
- 7 LoRaWAN concentrator HAT
- 8 microSD card, high endurance
- 9 LoRa antenna and SMA bulkhead
- 10 DC-DC converter, 12 V to 5 V
- 11 DIN UPS module
- 12 LiFePO4 backup pack, 12.8 V
- 13 Terminal blocks and fuse

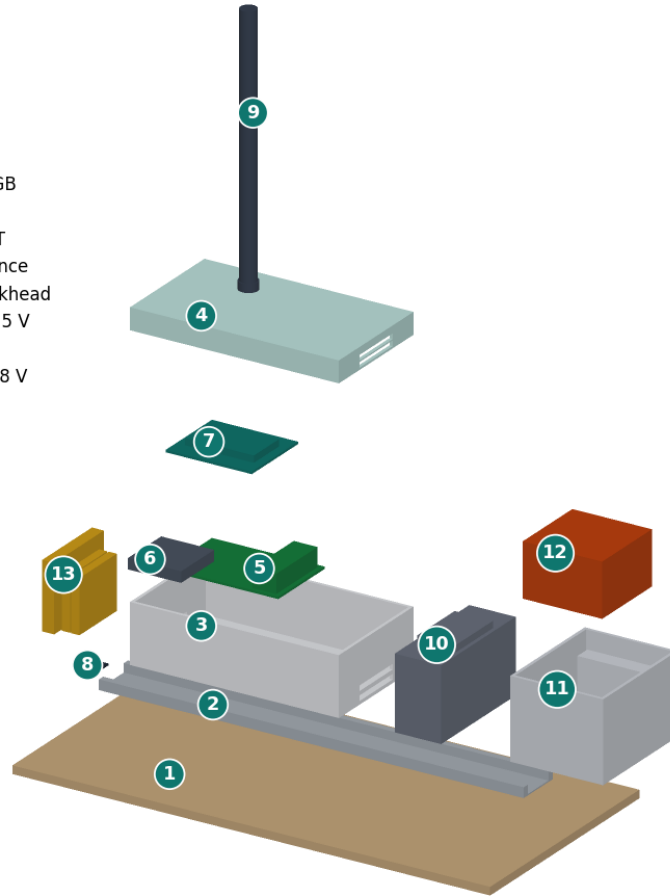


Figure 3. Exploded view with BOM numbers.

TwinKit: cutaway

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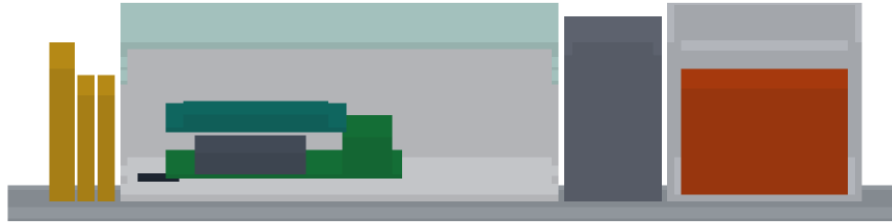


Figure 4. Cutaway through the gateway enclosure (left), converter (center) and UPS with backup pack (right).

■ Sizing results

The values below are from the TRL 3 calculation note TWK-CAL-001, which replaces the TRL 2 first-order estimates; tags in brackets point to lines of `docs/04-calcs/sizing.py`.

Table 2. Sizing results (TWK-CAL-001).

Quantity	Value	Basis	Requirement
Uplinks handled	600 per hour, 75 per channel [A1]	50 nodes x 12 readings per hour over 8 channels	R1
Time on air, 20-byte reading	71.9 ms at SF7, 246.8 ms at SF9, 1810.4 ms at SF12 [A2]	Semtech formula, 13 bytes of LoRaWAN overhead	R1
Uplink loss from collisions	0.30 % at SF7, 1.02 % at SF9, 7.27 % at SF12; 0.21 % for an SF7 to SF9 mix [A2, A3]	Pure ALOHA, all nodes on one spreading factor	R1 met with adaptive data rate (0.21 %) [A6]
Storage per year	4.73 GB uncompressed, 0.55 GB compressed [B1]	52.6 million values at 90 bytes	R3 met (56 GB free)
Card writes	35 to 53 GB per year, 5.5 to 8.2 card cycles [B4]	Write-ahead log per commit, assumed card amplification 10	R3, wear to watch
Reading to dashboard	0.9 to 4.7 s [C1]	Stage-by-stage latency budget	R5 met (60 s)
Average power	6.41 W, 0.53 A at 12 V [D1]	Board 4 W, fan 0.5 W, concentrator 1 W, converter 90 %, UPS 0.3 W	R8 met (8 W)
Peak input	20.6 W, 2.29 A at 9 V [D2]	Full load, downlink and pack charging	Fuse 3.15 A T [D3]
Backup time	2.40 h new, 1.63 h aged at 0 °C [E1]	15.36 Wh usable at 6.41 W	R9 met (30 min)
Processor at 40 °C ambient	69.1 °C light load, 96.9 °C sustained full load, vented [F2]	5 W/m ² K, 640 mm ² vents 42 mm apart, 4 K/W with fan	R12 met at light load; heavy jobs in cool hours [F6]
Memory	about 2.42 GB of 4 GB [G1]	Stack budget	Supports the 4 GB board
Rail length used	294 mm, 16.8 modules [H1]	From <code>cad/src/model.py</code>	R10 met (350 mm)
Parts cost	\$290.00 [I1]	Priced <code>bom/bom.csv</code> , indicative	R14 met (\$300 budget)

Thermal is still the main technical uncertainty. At light load the processor stays about 16 K below its throttle point at 40 °C, but a sustained heavy job would throttle, and doubling the vents barely helps (TWK-CAL-001 [F5]). Heavy database jobs therefore run in cool hours by design. A larger or metal enclosure is the hardware fallback if TRL 4 tests (on hold) show throttling.

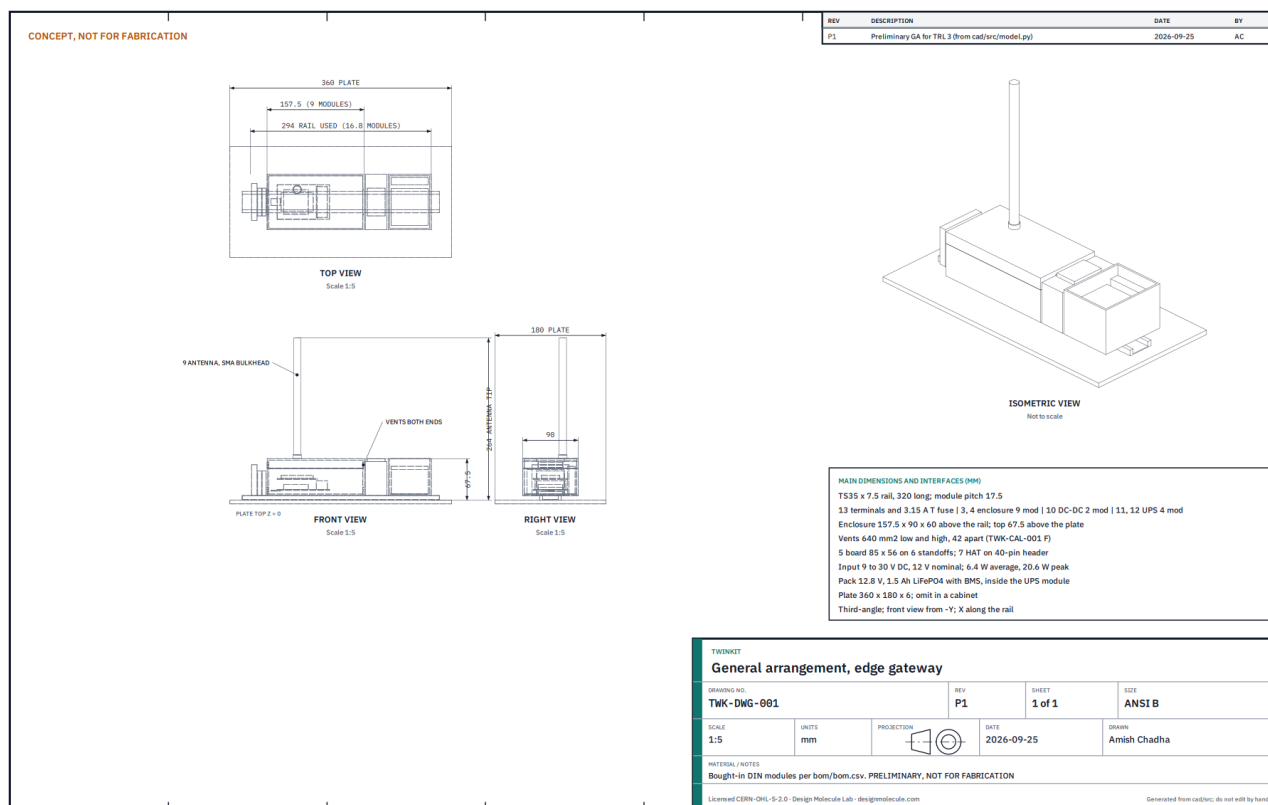


Figure 5. General arrangement TWK-DWG-001 Rev P1, generated from the parametric model `cad/src/model.py` (PDF). Preliminary, not for fabrication.

■ Key design choices

- **Compare with the calculation, not only with thresholds.** Most dashboards alarm on fixed limits. TwinKit compares each reading with what the design said it should be at that moment, so a battery that charges more slowly than the power budget predicted is caught before it goes flat.
- **Reuse the portfolio's own model outputs.** Every lab repo already exports a `model.glb` with named, BOM-numbered parts. The twin file points at those names, so no separate 3D model is built for the twin.
- **Edge first.** Everything runs on the gateway. An upstream server is optional, which keeps the kit usable where connectivity is poor and keeps data with its owner.
- **LoRaWAN rather than a single-channel receiver.** An 8-channel concentrator follows the open standard, matches FieldNode's LoRaWAN radio and accepts third-party sensors, at about \$55 more than a point-to-point receiver. Decided by Amish, 2026-09-25 (TWK-DDR-002).
- **Lightweight stack over a twin platform.** Frameworks such as Eclipse Ditto assume a server cluster; a single-board computer running a network server, MQTT, TimescaleDB and a small Python service fits in about 2.4 GB of 4 GB (TWK-CAL-001 [G1]). Decided by Amish, 2026-09-25 (TWK-DDR-002).
- **Monitoring only.** No control outputs, so a software fault cannot move a pump or switch a load.

■ Relation to other lab projects

- **FieldNode** is the primary data source and the first example twin, decided by Amish on 2026-09-25. Its nodes must accept LoRaWAN adaptive data rate commands (recorded as a cross-repo action in `docs/REVIEW.md`). FieldNode sends 20-byte LoRaWAN uplinks every 15 min by default; TwinKit is sized for 5 min, which is conservative.
- **CityTwin** builds its public map and kiosk on TwinKit's gateway and software. Its review recommends costing the gateway here, in TwinKit. A gateway in a sealed street cabinet in sun is hotter than the 40 °C case sized here.
- **CellGuard** could protect a larger backup pack in a later version; at this size an off-the-shelf pack with a built-in BMS was decided by Amish on 2026-09-25.
- **CalRig** calibration records could feed each channel's tolerance in the twin file.

■ Safety

Safety: The backup pack is a lithium (LiFePO₄) battery. Use a pack with a built-in BMS and fuse, charge it only through the UPS module within the maker's limits, keep it away from heat, and never leave a first build charging unattended.

Safety: The gateway runs from 12 V DC. Use a certified, double-insulated mains adapter; any wiring to an existing 12 V system or into a mains cabinet must be done or checked by a qualified electrician under local code.

Safety: A network gateway is an attack surface. Change every default credential, keep it off the public internet until it is hardened, apply updates, and treat sensor data as the owner's data.

Safety: TwinKit is a monitoring aid. It must not be relied on as a safety system, and it does not replace alarms, protection devices or inspections that a regulation or the equipment maker requires.

■ Open questions after TRL 3

- Co-design partner outside the lab. Proposed, awaiting Amish.
- Processor temperature, card endurance and the power figures are assumptions that only measurement can settle.

Concept media: [blueprint sheet](#), [interactive 3D model](#).